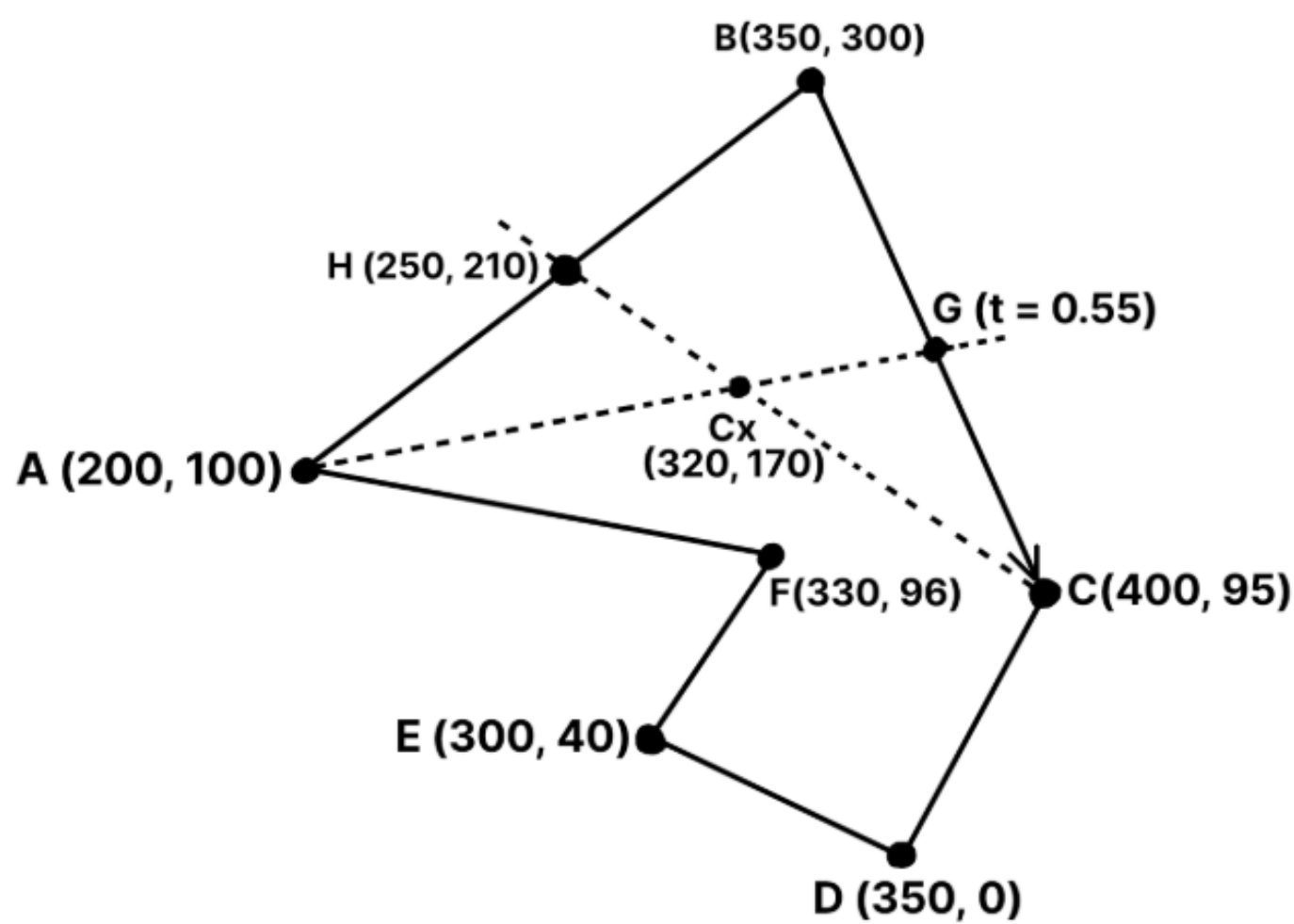




The RGB colors are given as follows:

$$A = (0.5, 0.7, 0.3), B = (0.3, 0.9, 0.8), C = (0.6, 0.4, 0.2),$$

$$D = (0.7, 0.8, 0.1), E = (0.3, 0.5, 0.7), F = (0.5, 0.7, 0.9)$$

Using Gouraud shading, compute the interpolated color at point $C_x(320, 170)$ using:

- Vertex A and Vertex G
- Vertex C and Vertex H

$$\begin{aligned} \text{a) } \text{Color}_G &= \text{Color}_B + t(\text{Color}_C - \text{Color}_B) \\ &= (0.3, 0.9, 0.8) + 0.55(0.6 - 0.3, 0.4 - 0.9, 0.2 - 0.8) \\ &= (0.465, 0.675, 0.47) \end{aligned}$$

$$\begin{aligned} G &= B + t(C - B) \\ &= (350, 300) + 0.55(400 - 350, 95 - 300) \\ &= (377.5, 187.25) \rightarrow \text{Coordinate} \end{aligned}$$

$$\begin{aligned} \text{Color}_{C_x} &= \text{Color}_A + \frac{d_{AC_x}}{d_{AG}} (\text{Color}_G - \text{Color}_A) \\ &= (0.5, 0.7, 0.3) + \frac{320 - 200}{377.5 - 200} (0.465 - 0.5, 0.675 - 0.7, 0.47 - 0.3) \\ &= (0.476, 0.683, 0.415) \end{aligned}$$

$$\begin{aligned} dx &= 377.5 - 200 \\ &= 177.5 \\ dy &= 187.25 - 100 \\ &= 87.25 \\ |dx| &> |dy| \end{aligned}$$

$$\begin{aligned} \text{b) } \text{Color}_H &= \text{Color}_A + \frac{d_{AH}}{d_{AB}} (\text{Color}_B - \text{Color}_A) \\ &= (0.5, 0.7, 0.3) + \frac{210 - 100}{300 - 100} (0.3 - 0.5, 0.9 - 0.7, 0.8 - 0.3) \\ &= (0.39, 0.81, 0.575) \end{aligned}$$

$$\begin{aligned} dx &= 350 - 200 \\ &= 150 \\ dy &= 300 - 100 \\ &= 200 \\ |dy| &> |dx| \end{aligned}$$

$$\begin{aligned} \text{Color}_{C_x} &= \text{Color}_C + \frac{d_{CC_x}}{d_{CH}} (\text{Color}_H - \text{Color}_C) \\ &= (0.6, 0.4, 0.2) + \frac{320 - 400}{250 - 400} (0.39 - 0.6, 0.81 - 0.4, 0.575 - 0.2) \\ &= (0.488, 0.619, 0.400) \end{aligned}$$

$$\begin{aligned} dx &= 250 - 400 \\ &= -150 \\ dy &= 210 - 95 \\ &= 115 \\ |dx| &> |dy| \end{aligned}$$